Student Name: _____

Class Name: **ALEKS Integrated Mathematics II**

Number of Questions: **52**

Instructor Name: **Bodanske, Daniel**

Question 1 of 52

Rewrite the equation in $Ax+By=C$ form.
Use integers for A, B, and C.

$$y-6=5(x-3)$$

Question 2 of 52

Find the slope and the y-intercept of the line.

$$3x-y=1$$

Write your answers in simplest form.

Question 3 of 52

A table of values of a linear function is shown below.

x	0	1	2	3	4
y	8	12	16	20	24

Find the slope and y-intercept of the function's graph, and find the equation for the function.

slope: _____

y-intercept: _____

equation: _____

Question 4 of 52

A line passes through the point $(6,\ 2)$ and has a slope of $\frac{5}{2}$.

Write an equation in point-slope form for this line.

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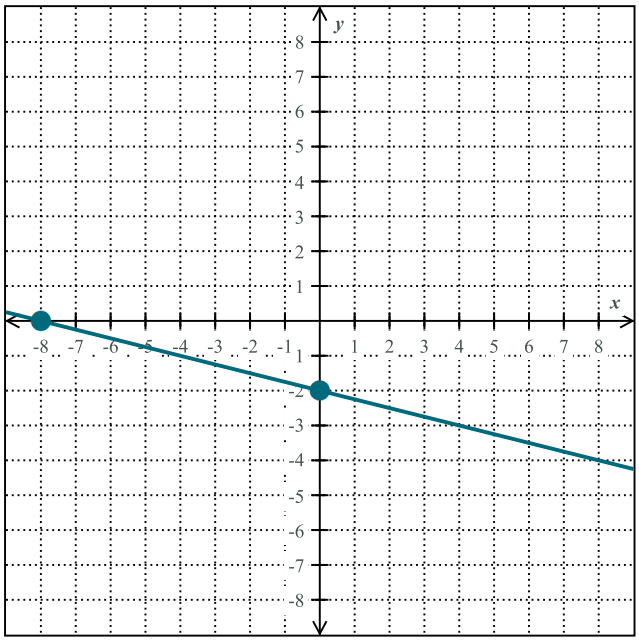
Question 5 of 52

Write an equation of the line below.



Question 6 of 52

Find an equation for the line that passes through the points $(-4,\ 2)$ and $(2,\ -2)$.

Question 7 of 52

Chau has 26 shirts to fold.

Let N be the number of shirts he would have left to fold after folding F of them.

Write an equation relating N to F. Then use this equation to find the number of shirts he would have left to fold after folding 23 of them.

Equation:
Number left to fold after folding 23 of them: _____ shirts

Question 8 of 52

Eric is a software salesman. His base salary is $\$1700$, and he makes an additional $\$40$ for every copy of *History is Fun* he sells.

Let P represent his total pay (in dollars), and let N represent the number of copies of *History is Fun* he sells. Write an equation relating P to N. Then use this equation to find his total pay if he sells 27 copies of *History is Fun*.

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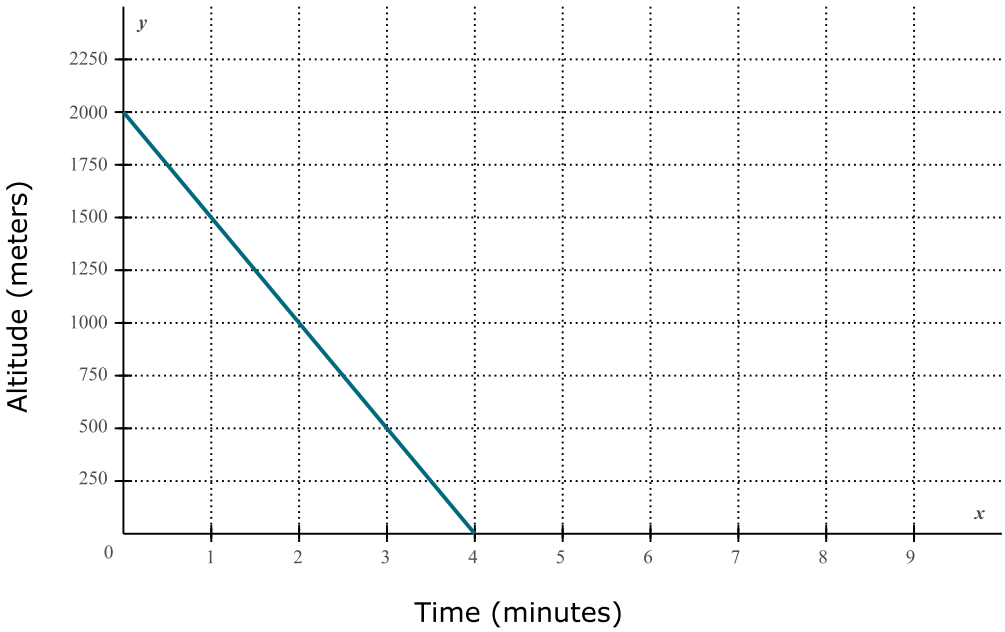
Question 9 of 52

Brian is riding a gondola from the top of a ski area down to the lodge. The graph below shows the linear relationship between the time x (in minutes) since the gondola left the ski area and the altitude y (in meters) of the gondola.



Write a linear equation to represent the relationship between x and y.

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Question 10 of 52

A researcher is heating a metal. He heats the metal so that the temperature of the metal rises at a constant rate. After 29 minutes of heating, the metal is 526 °C. After 41 minutes, the metal is 730 °C.

(a) Choose the statement that best describes how the time and the temperature of the metal are related. Then fill in the blank.

- As time increases, the temperature of the metal decreases.
  The temperature of the metal decreases at a rate of _____ °C per minute.

- As time increases, the temperature of the metal increases.
  The temperature of the metal increases at a rate of _____ °C per minute.

(b) What was the temperature of the metal when the researcher started heating it?

_____ °C

Question 11 of 52

Jina is putting money into a checking account. Let y represent the total amount of money in the account (in dollars). Let x represent the number of weeks Jina has been adding money. Suppose that x and y are related by the equation $650 + 20x = y$.

Answer the questions below.

Note that a change can be an increase or a decrease.
For an increase, use a positive number. For a decrease, use a negative number.

(a) What was the starting amount of money in the account?

$_____

(b) What is the change per week in the amount of money in the account?

$_____

Question 12 of 52

The first three terms of an arithmetic sequence are as follows.

$5, 8, 11$

Find the next two terms of this sequence.

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Question 13 of 52

The first three terms of an arithmetic sequence are as follows.

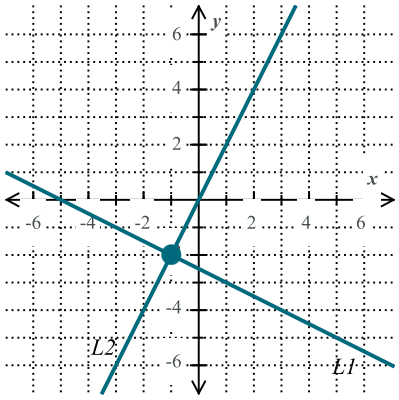
$$-12,\ -19,\ -26$$

Find the next two terms of this sequence.

Question 14 of 52

Three systems of linear equations are shown.
For each system, choose the best description of its solution.
If the system has exactly one solution, give its solution.

(a)
$$\begin{cases} \text{Line } 1: & y=-\frac{1}{2}x-\frac{5}{2} \\ \text{Line } 2: & y=2x \end{cases}$$

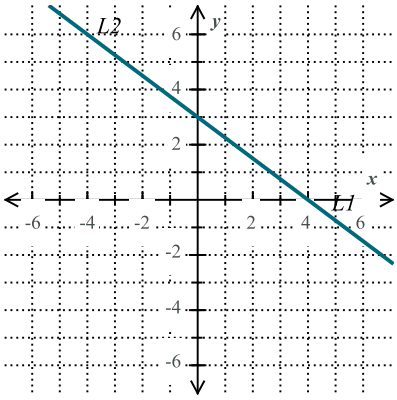


Describe the solution.

- The system has exactly one solution.

  Solution: (___ , ___)
- The system has infinitely many solutions.
- The system has no solution.

(b)
$$\begin{cases} \text{Line } 1: & y=-\frac{3}{4}x+3 \\ \text{Line } 2: & 3x+4y=12 \end{cases}$$

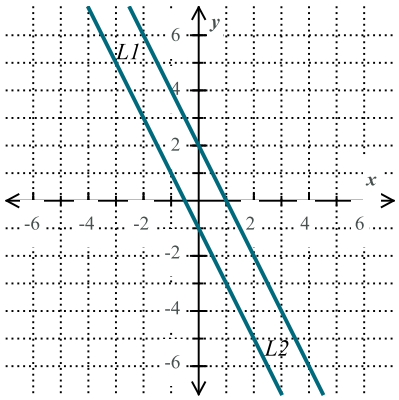


Describe the solution.

- The system has exactly one solution.

  Solution: (___ , ___)
- The system has infinitely many solutions.
- The system has no solution.

(c)
$$\begin{cases} \text{Line } 1: & y=-2x-1 \\ \text{Line } 2: & y=-2x+2 \end{cases}$$



Describe the solution.

- The system has exactly one solution.

  Solution: (___ , ___)
- The system has infinitely many solutions.
- The system has no solution.

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Question 15 of 52

For each system of linear equations shown below, classify the system as "consistent dependent," "consistent independent," or "inconsistent." Then, choose the best description of its solution. If the system has exactly one solution, give its solution.

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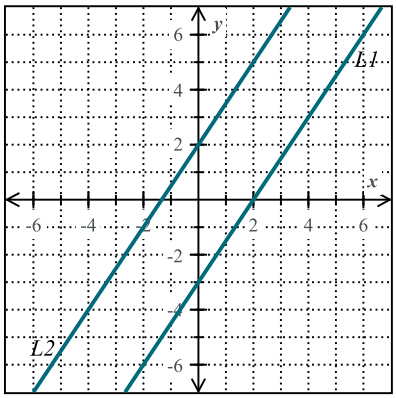
System A

L1: $y = \frac{3}{2}x - 3$

L2: $y = \frac{3}{2}x + 2$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

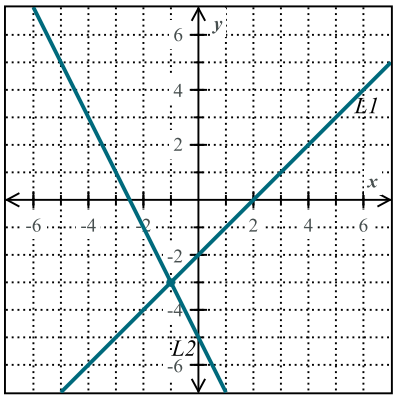
- a unique solution: - no solution - infinitely many solutions

Solution: (____ , ____)

System B

L1: $y = x - 2$

L2: $y = -2x - 5$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

- a unique solution: - no solution - infinitely many solutions

Solution: (____ , ____)

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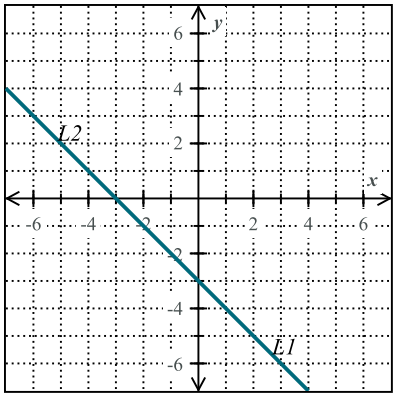
System C

L1: $y = -x - 3$

L2: $x + y = -3$



This system of equations is:

- consistent dependent - consistent independent - inconsistent

This means the system has:

- a unique solution: - no solution - infinitely many solutions

Solution: (___ , ___)

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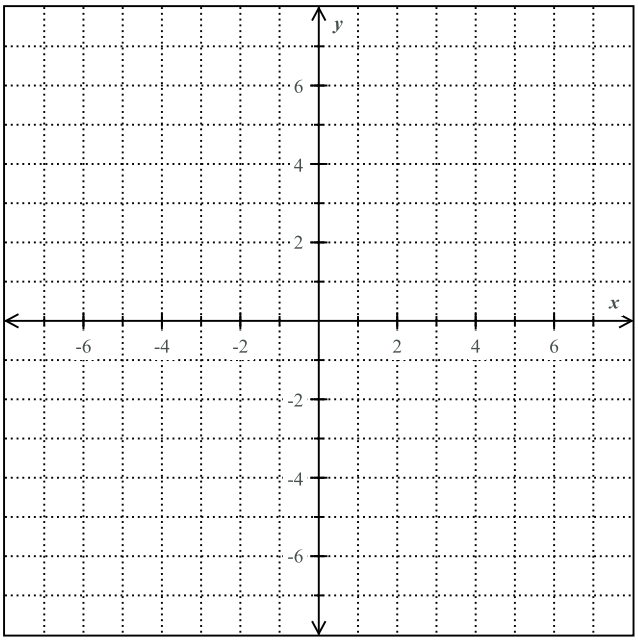
Question 16 of 52

Here is a system of equations.

$$\begin{cases} y = -3x + 5 \\ y = -3x + 6 \end{cases}$$

Graph the system. Then write its solution. Note that you can also answer "No solution" or "Infinitely many" solutions.



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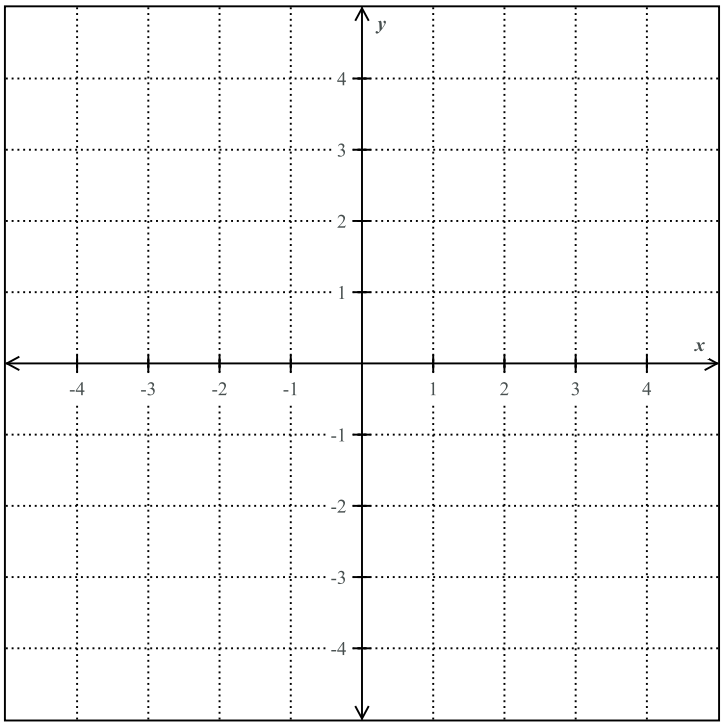
Question 17 of 52

Here is a system of equations.

$$\begin{cases} y = 3x - 3 \\ y = -\dfrac{1}{4}x + 1 \end{cases}$$

(a) Graph the system.



(b) Use your graph to estimate the solution to the system. Write the coordinates to the tenths place.

(_____, _____)

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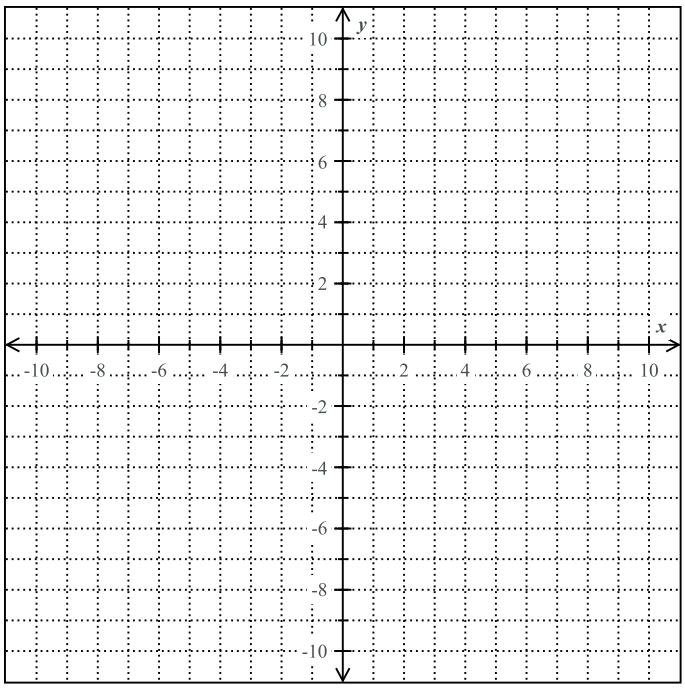
Question 18 of 52

Graph the system below and write its solution.

$$\begin{cases} y = \frac{1}{2}x - 1 \\ 2x + y = 4 \end{cases}$$

Note that you can also answer "No solution" or "Infinitely many" solutions.



Solution: _____

Question 19 of 52

Use a graphing calculator to solve the system of equations.

$$y = -2x + 1$$
$$y = -11x - 3$$

Round to the nearest hundredth.

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Question 20 of 52

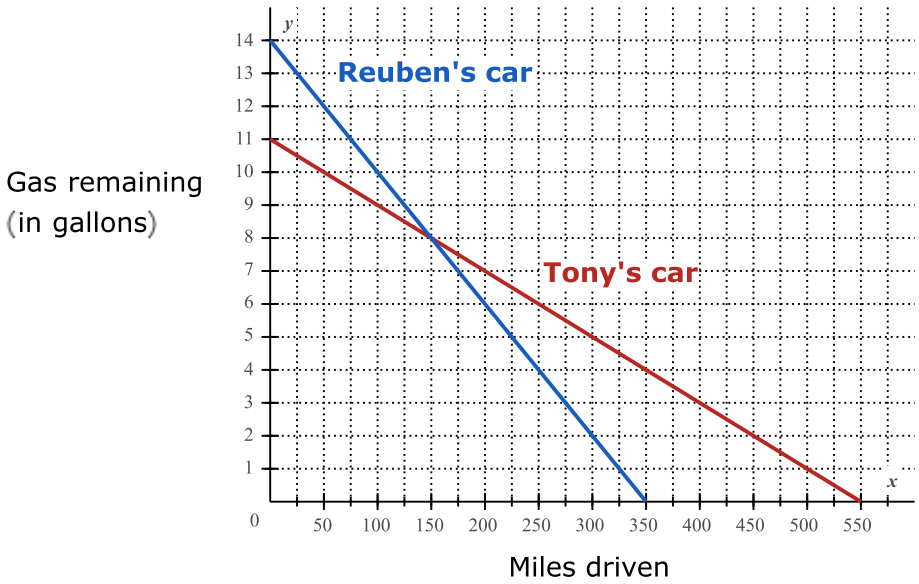
Use a graphing calculator to solve the system of equations.

$$5 - 0.7x = 0.2y$$
$$-y + 0.4x = 1.8$$

Round to the nearest hundredth.

Question 21 of 52

Tony and Reuben are driving separate cars to Boston. They begin the trip with full gas tanks. The amount of gas (in gallons) remaining in the tank of each car depends on the number of miles driven, as shown below.



(a) After 250 miles driven, which car will have less gas remaining?

- Tony's car
- Reuben's car

How much less gas will it have?

_____ gallon(s)

(b) After how many miles will the tanks contain the same amount of gas?

_____

If the number of miles driven is less than this, which car will have more gas remaining?

- Tony's car
- Reuben's car

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Question 22 of 52

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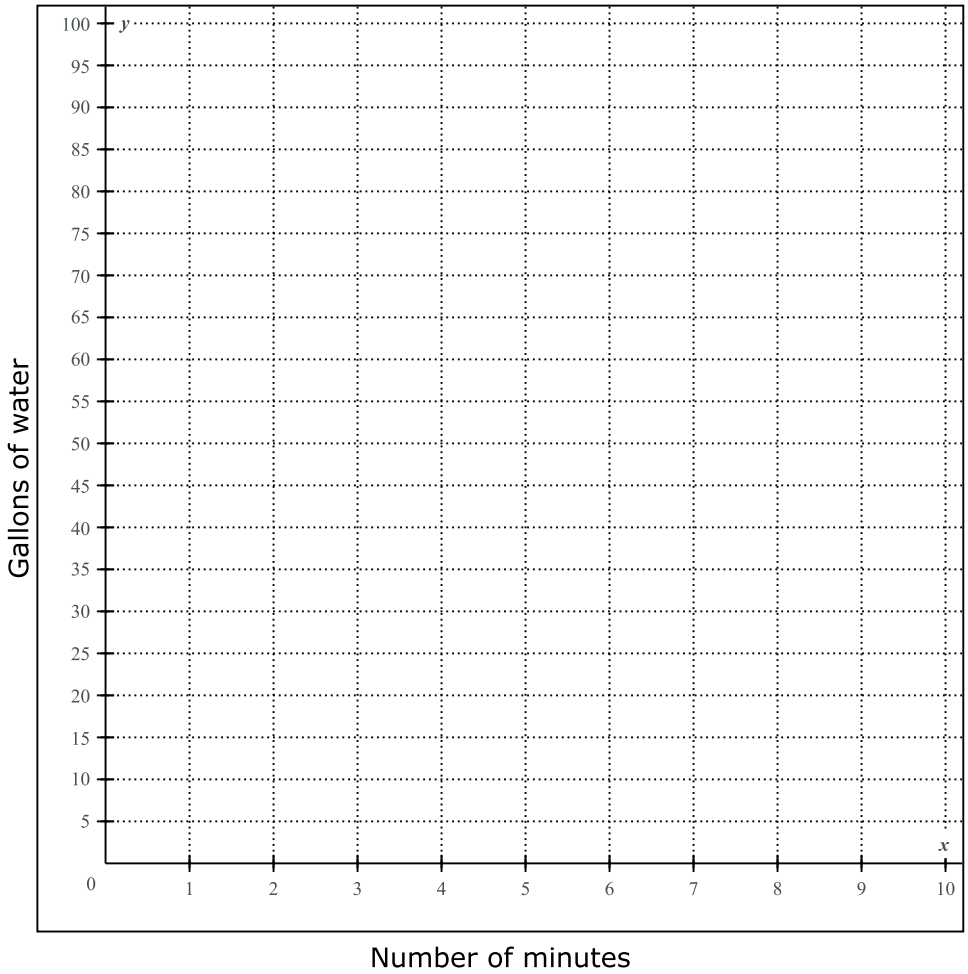
Teresa works at a pet store and is cleaning two fish tanks. Tank A has 80 gallons of water and Tank B has 35 gallons of water. With two pumps running at the same time, Teresa drains 30 gallons per minute from Tank A and adds 10 gallons of water per minute to Tank B.

Using x for the number of minutes that have passed, the amount of water y (in gallons) in each tank is given by the system of equations.

Tank A: $y = 80 - 30x$

Tank B: $y = 35 + 10x$

(a) Graph the system of equations.



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(b) Using the graph, about how many minutes will pass before the tanks have the same amount of water? Write your answer to the tenths place.

_____ minutes

(c) Using the graph, about how many gallons of water will be in each tank when they have the same amount of water? Write your answer as a whole number.

_____ gallons

Question 23 of 52

For each ordered pair, determine whether it is a solution to the system of equations.

$$\begin{cases} -4x + 5y = 6 \\ 7x - 3y = 1 \end{cases}$$

(a) Is $(-2, -5)$ a solution?

Yes No

(b) Is $(1, 2)$ a solution?

Yes No

(c) Is $(-9, -6)$ a solution?

Yes No

(d) Is $(0, -8)$ a solution?

Yes No

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Question 24 of 52

Use substitution to solve the system.

$$\begin{cases} 3x + 4y = 8 \\ 2x - 9 = y \end{cases}$$

$x =$ _____

$y =$ _____

Question 25 of 52

Solve the following system of equations.

$$\begin{aligned} 2x + 4y &= -8 \\ -2x + 3y &= 29 \end{aligned}$$

Question 26 of 52

Solve the following system of equations.

$$\begin{aligned} -5x + 7y &= -18 \\ -2x - 9y &= -19 \end{aligned}$$

Question 27 of 52

Solve the following system of equations.

$$-\frac{1}{5}x + \frac{1}{3}y = -2$$

$$\frac{5}{6}x - \frac{1}{2}y = -5$$

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Question 28 of 52

Solve the following system of equations.

$$0.16x - 0.07y = 0.77$$
$$-0.4x + 1.5y = 4.7$$

Question 29 of 52

Two systems of equations are given below.
For each system, choose the best description of its solution.
If applicable, give the solution.

System A $y = 9x + 7$ $y = 9x + 4$	○ The system has no solution. ○ The system has a unique solution: $(x, y) = (_____, _____)$ ○ The system has infinitely many solutions.
System B $-x - 2y = 3$ $x + 2y = -3$	○ The system has no solution. ○ The system has a unique solution: $(x, y) = (_____, _____)$ ○ The system has infinitely many solutions.

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Question 30 of 52

Two systems of equations are given below.
For each system, choose the best description of its solution.
If applicable, give the solution.

(a) Solve the following system of equations.

$$\begin{cases} -3x + y = -6 \\ 3x - y = 6 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\square, \square)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \square$

(b) Solve the following system of equations.

$$\begin{cases} -2x + y = 4 \\ 2x - y = 4 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (\square, \square)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y = \square$

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(a) Solve the following system of equations.

$$\begin{cases} -3x + y = -6 \\ 3x - y = 6 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (____, ____)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y =$ _____

(b) Solve the following system of equations.

$$\begin{cases} -2x + y = 4 \\ 2x - y = 4 \end{cases}$$

- The system has no solution.
- The system has a unique solution.

  $(x, y) = (____, ____)$

- The system has infinitely many solutions. They must satisfy the following equation.

  $y =$ _____

Question 31 of 52

A textbook store sold a combined total of 445 chemistry and biology textbooks in a week. The number of biology textbooks sold was 87 less than the number of chemistry textbooks sold. How many textbooks of each type were sold?

Question 32 of 52

A party rental company has chairs and tables for rent. The total cost to rent 2 chairs and 6 tables is $\$52$. The total cost to rent 5 chairs and 3 tables is $\$31$. What is the cost to rent each chair and each table?

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Question 33 of 52

A fruit company packages its fruit into two types of boxes: large and small. This morning, the company made two deliveries. The table below shows the number of boxes in each delivery and the total weight (in kilograms).

	First delivery	**Second delivery**
Number of large boxes	3	8
Number of small boxes	2	4
Total weight (in kilograms)	53	133

Let x be the weight (in kilograms) of each large box.

Let y be the weight (in kilograms) of each small box.

(a) Write a system of equations that could be used to find the weight (in kilograms) of each type of box.

___ x + ___ y = ____

___ x + ___ y = ____

(b) How much does each type of box weigh (in kilograms)?

Weight of each large box: _____ **kilograms**

Weight of each small box: _____ **kilograms**

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Question 34 of 52

Last month Mary downloaded 5 games and 3 audiobooks on her tablet for a total of $\$32$. This month she downloaded 2 games and 12 audiobooks on her tablet for a total of $\$83$. Each game costs the same, and each audiobook costs the same.

Let x be the cost (in dollars) of downloading a game.
Let y be the cost (in dollars) of downloading an audiobook.

(a) Write a system of equations that could be used to find the cost (in dollars) for each game and each audiobook.

_____ x + _____ y = _____

_____ x + _____ y = _____

(b) How much did it cost (in dollars) to download each game and each audiobook?

Cost for each game: $\$$ _____

Cost for each audiobook: $\$$ _____

Question 35 of 52

Cars produce carbon dioxide pollution. The rate they produce the pollution is measured in $\frac{\text{g}}{\text{km}}$ (grams per kilometer). Suppose that a family's two cars produce a combined rate of $75\ \frac{\text{g}}{\text{km}}$ of carbon dioxide pollution. During one day last week, the first car drove 40 kilometers and the second drove 15 kilometers. The two cars produced a combined total of 2125 grams of carbon dioxide pollution.

At what rate does each car produce carbon dioxide pollution?

First car: _____ $\frac{\text{g}}{\text{km}}$

Second car: _____ $\frac{\text{g}}{\text{km}}$

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Question 36 of 52

A chef is going to use a mixture of two brands of Italian dressing. The first brand contains 7% vinegar, and the second brand contains 12% vinegar. The chef wants to make 240 milliliters of a dressing that is 11% vinegar.

How much of each brand should she use?

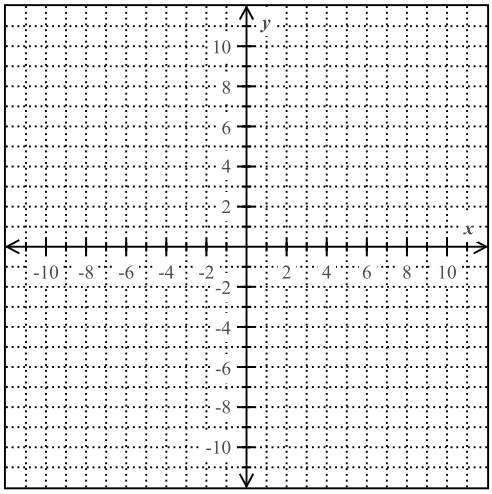
First brand: _____ milliliters

Second brand: _____ milliliters

Question 37 of 52

Graph the solution to the following system of inequalities.

$$y < 3x + 2$$
$$y > -2x + 3$$



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Question 38 of 52

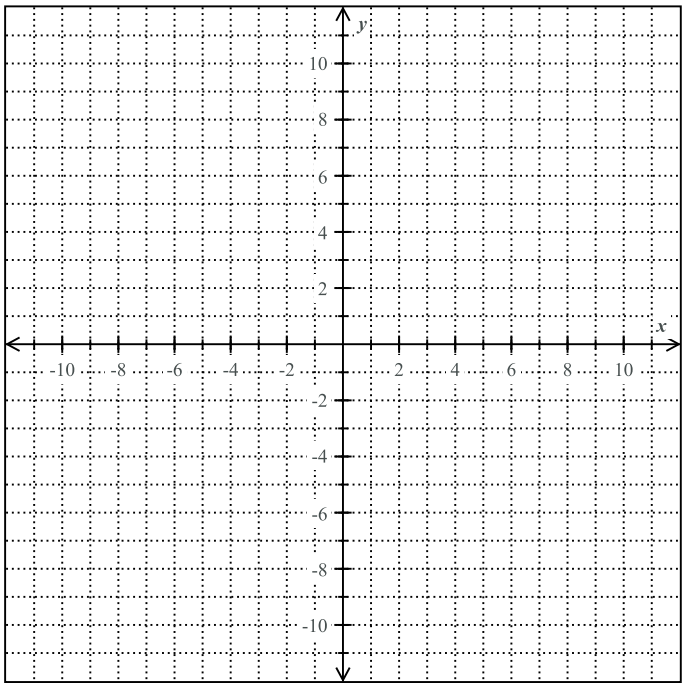
(a) Graph the solution to the following system of inequalities.

$$\begin{cases} 7x + 5y < 20 \\ -6x + 5y \leq 15 \end{cases}$$

(b) Give the coordinates of one point in the solution set.

(a) Graph the solution to the following system of inequalities.

$$\begin{cases} 7x + 5y < 20 \\ -6x + 5y \leq 15 \end{cases}$$



(b) Give the coordinates of one point in the solution set.

Point in the solution set: (____ , ____)

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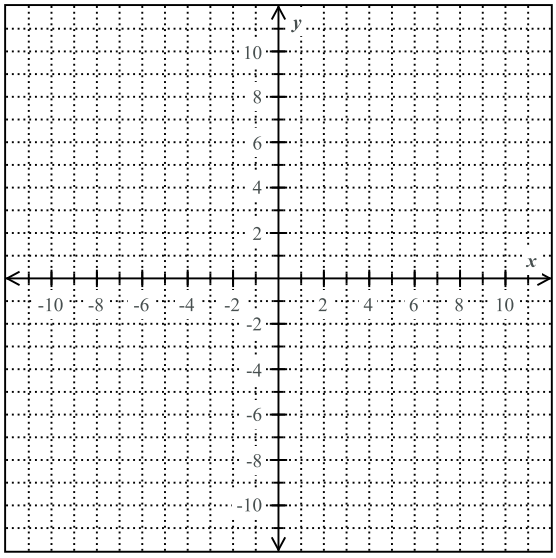
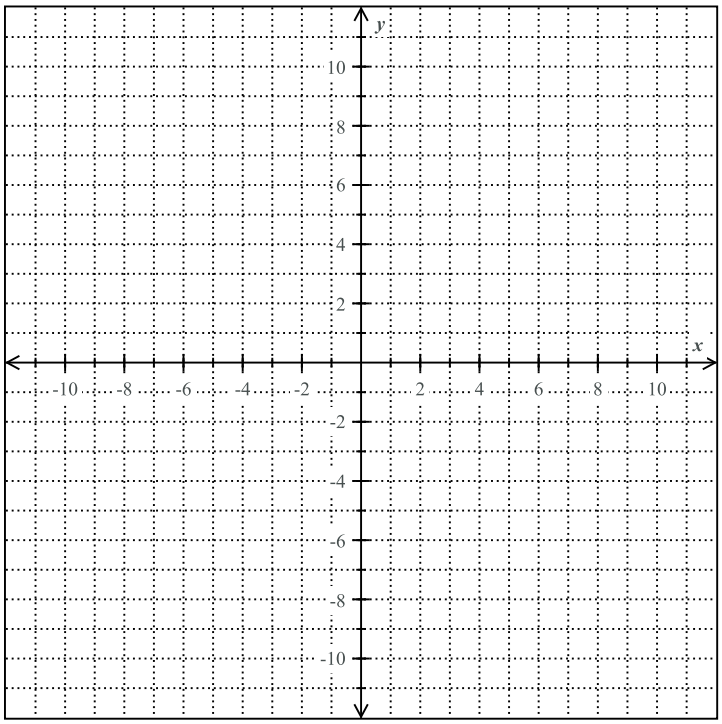
Question 39 of 52

Graph the solution to the following system of inequalities.

$$\begin{cases} 7x + 5y > 35 \\ x \geq 0 \\ y \geq 0 \end{cases}$$





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Question 40 of 52

Scott's final exam has true/false questions and multiple choice questions. The table below shows how many points each of these questions is worth.

	True/False	Multiple choice
Number of points each question is worth	3	4

Let x be the number of true/false questions he gets correct.

Let y be the number of multiple choice questions he gets correct.

Scott needs more than 93 points on the exam to get an A in the class. Using the values and variables given, write an inequality describing this.

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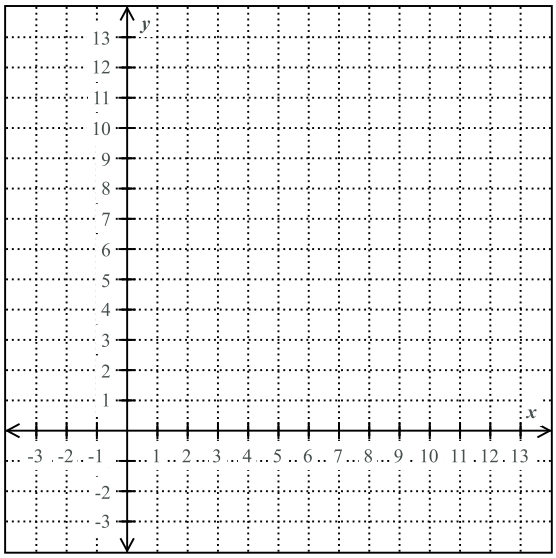
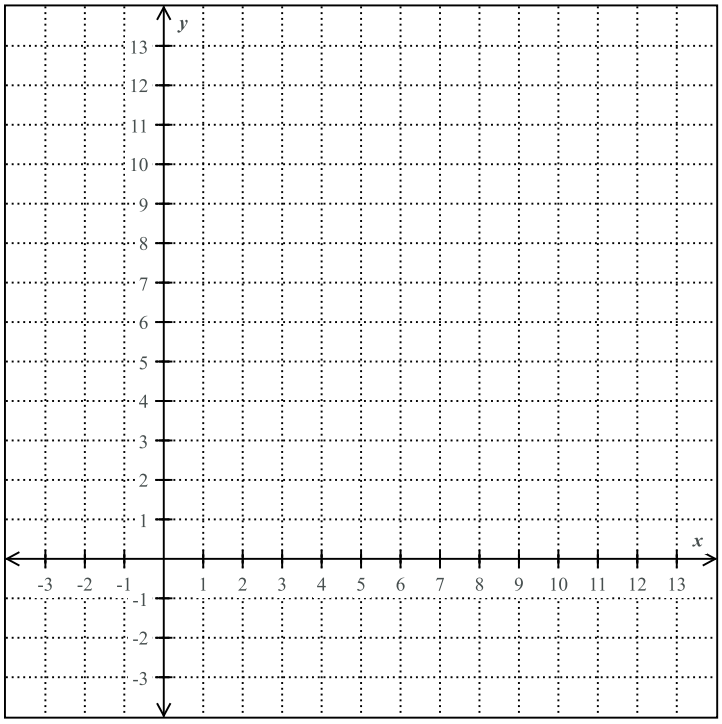
Question 41 of 52

Ravi does a weekly exercise program consisting of cardiovascular work and weight training. Each week, he exercises for at least 10 hours. He spends at most 12 hours doing cardiovascular work. He spends at most 5 hours on weight training. Let x denote the time (in hours) that Ravi spends doing cardiovascular work. Let y denote the time (in hours) that he spends on weight training. Shade the region corresponding to all values of x and y that satisfy these requirements.





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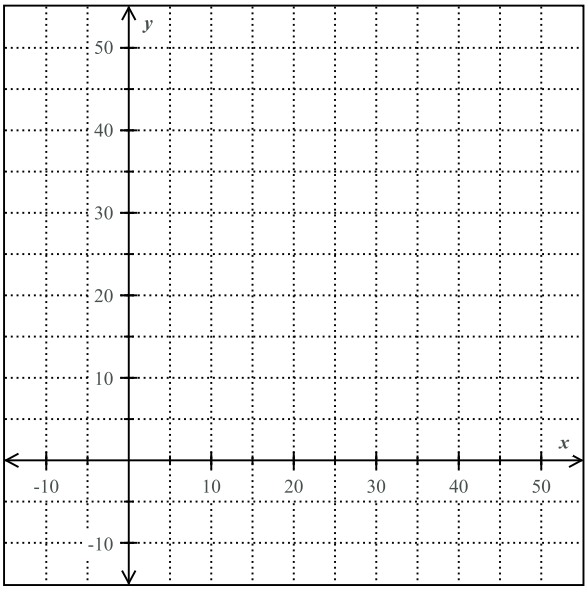
Question 42 of 52

A store has two types of animal feed available. Type A contains 5 pounds of oats and 8 pounds of corn per bag. Type B contains 7 pounds of oats and 3 pounds of corn per bag. A farmer buys both types and combines them so that the resulting mixture has at least 175 pounds of oats and at least 120 pounds of corn. The store only has 30 bags of type A feed and 25 bags of type B feed in stock. Let x be the number of type A bags purchased. Let y be the number of type B bags purchased. Shade the region corresponding to all values of x and y that satisfy these requirements.



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Question 43 of 52

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A magician is setting up a magic ball trick. She will be using some number of black balls and some number of white balls, placed in a bag. Either of those numbers could be zero. Let x be the number of black balls she uses. Let y be the number of white balls she uses. There will be no other balls used.

Answer the parts below.

(a) Each black ball weighs 30 grams and each white ball weighs 60 grams. For the trick to work, the magician needs the total weight of the balls to be at most 600 grams. Using those values and the variables given, write an inequality describing this constraint.

(b) The magician needs at least 11 total balls to perform the trick. Using the values and the variables given, write an inequality describing this constraint.

(c) From the choices below, choose the correct additional constraints on x and y.

- $x > y$ because there will be more white balls than black balls in the bag.
- $x \geq 0$ and $y \geq 0$ because the number of black balls and number of white balls cannot be negative.
- $x < 0$ and $y < 0$ because each ball takes up space in the bag.

(d) Suppose that the constraints in parts (a) through (c) apply. For each description below, decide whether it gives possible numbers of black balls and white balls that could be used. Note, you may use the Scratch Area below the table, but the graph will not be part of the answer.

Description	Possible?
8 black balls and 4 white balls	Yes No
6 black balls and 10 white balls	Yes No
12 black balls and 0 white balls	Yes No

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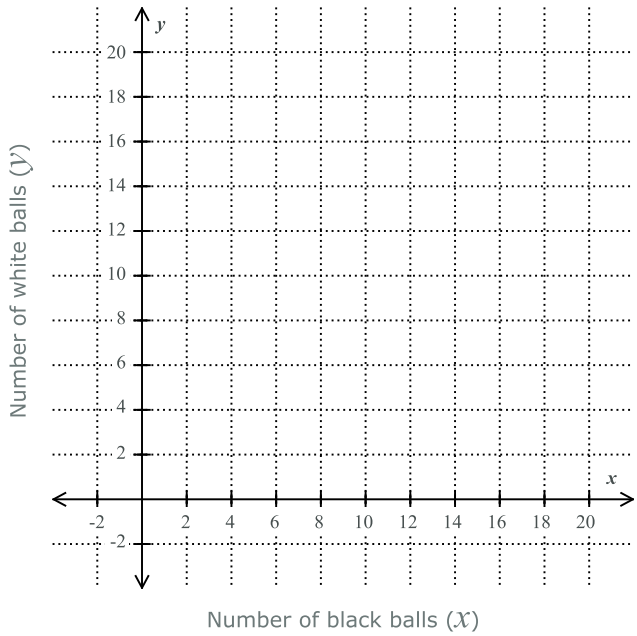
Scratch Area (Not Part of Answer)



Question 44 of 52

Find the maximum value of the function $z = 3x + 9y$ subject to the following constraints.

$$\begin{cases} 6x + 7y \leq 98 \\ 4x + 3y \leq 52 \\ x \geq 0 \\ y \geq 0 \end{cases}$$

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Question 45 of 52

A mining company owns two mines. These mines produce an ore that can be graded into two classes: regular grade and low grade. The company must produce at least 420 tons of regular-grade and 540 tons of low-grade ore per week. The first mine produces 6 tons of regular-grade and 18 tons of low-grade ore per hour. The second mine produces 18 tons of regular-grade and 9 tons of low-grade ore per hour. The operating cost of the first mine is $\$7000$ per hour, and the operating cost of the second mine is $\$22,000$ per hour. The first mine can be operated no more than 37 hours a week, and the second mine can be operated no more than 28 hours a week. How many hours per week should each mine be operated to minimize the cost?

Note that the ALEKS graphing calculator can be used to make computations easier.

Question 46 of 52

The sum of three numbers is 137. The second number is 5 more than the first. The third number is 4 times the first. What are the numbers?

Question 47 of 52

Solve the following system.

$$\begin{cases} z = 1 \\ -2y + z = 3 \\ 5x + 3y - 2z = 5 \end{cases}$$

Question 48 of 52

Solve the system.

$$\begin{aligned} -x + y + 2z &= -3 \\ 3x - y + z &= -2 \\ 2x - 3y - 4z &= 1 \end{aligned}$$

$x =$

$y =$

$z =$

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Question 49 of 52

Solve the following system.

$$\begin{cases} x - 4y + z = -16 \\ y - 3z = 9 \\ 3y - 2z = 13 \end{cases}$$

Question 50 of 52

Over the last 3 evenings, Lisa received a total of 87 phone calls at the call center. The first evening, she received 8 more calls than the second evening. The third evening, she received 3 times as many calls as the first evening.

How many phone calls did she receive each evening?

Number of phone calls the first evening: _____

Number of phone calls the second evening: _____

Number of phone calls the third evening: _____

Question 51 of 52

The Alexander family takes their clothes to the local cleaners. The table below gives the number of coats, shirts, and dresses they had cleaned over the course of three months. They had no other items cleaned there.

Number cleaned			
	January	February	March
Coats	5	9	8
Shirts	8	8	4
Dresses	4	7	5

If they spent $\$57$ at the cleaners in January, $\$89$ in February, and $\$68$ in March, what is the cost to have each item cleaned?

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Question 52 of 52

A table of values of a linear function is shown below.

x	-2	-1	0	1	2
y	-3	-4	-5	-6	-7

Find the slope and y-intercept of the function's graph, and find the equation for the function.

slope: _____

y-intercept: _____

equation: _____

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Class Name: **ALEKS Integrated Mathematics II** Number of Questions: **52**

Question 1 of 52

$-5x+y=-9$

Question 2 of 52

slope: 3
y-intercept: -1

Question 3 of 52

slope: 4

y-intercept: 8

equation: $y=4x+8$

Question 4 of 52

$y-2=\frac{5}{2}(x-6)$

Question 5 of 52

$y=-\frac{1}{4}x-2$

Question 6 of 52

$y=-\frac{2}{3}x-\frac{2}{3}$

Note: Other correct answers include $y-2=-\frac{2}{3}(x+4)$ and $2x+3y=-2$.

Question 7 of 52

Equation: $N=26-F$
Number left to fold after folding 23 of them: 3 shirts

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Question 8 of 52

Equation: $P = 1700 + 40N$

Total pay if Eric sells 27 copies: $\$2780$

Question 9 of 52

$y = -500x + 2000$

Question 10 of 52

(a) ○ As time increases, the temperature of the metal decreases.
The temperature of the metal decreases at a rate of $\square$ °C per minute.

◉ As time increases, the temperature of the metal increases.
The temperature of the metal increases at a rate of 17 °C per minute.

(b) 33 °C

Question 11 of 52

(a) What was the starting amount of money in the account?

$\$650$

(b) What is the change per week in the amount of money in the account?

$\$20$

Question 12 of 52

$5, 8, 11, 14, 17$

Question 13 of 52

$-12,\ -19,\ -26,\ -33,\ -40$

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Question 14 of 52

(a) ◉ The system has exactly one solution.

Solution: $(-1, -2)$

○ The system has infinitely many solutions.

○ The system has no solution.

(b) ○ The system has exactly one solution.

Solution: $(\square, \square)$

◉ The system has infinitely many solutions.

○ The system has no solution.

(c) ○ The system has exactly one solution.

Solution: $(\square, \square)$

○ The system has infinitely many solutions.

◉ The system has no solution.

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Question 15 of 52

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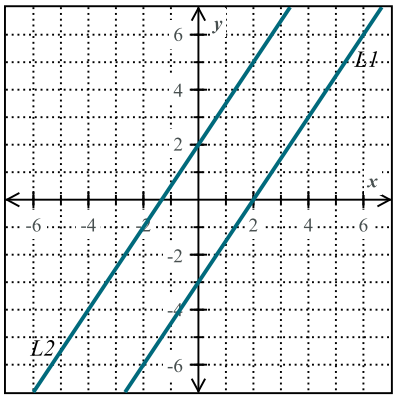
System A

L1: $y = \frac{3}{2}x - 3$

L2: $y = \frac{3}{2}x + 2$



This system of equations is:

- inconsistent

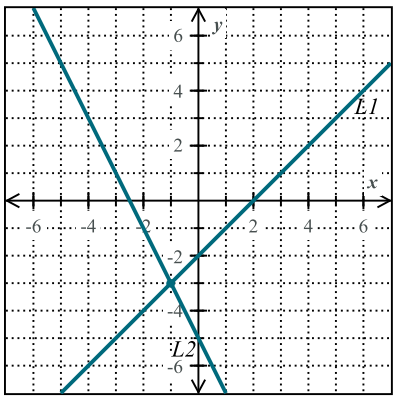
This means the system has:

- no solution

System B

L1: $y = x - 2$

L2: $y = -2x - 5$



This system of equations is:

- consistent independent

This means the system has:

- a unique solution:

Solution: $(-1, -3)$

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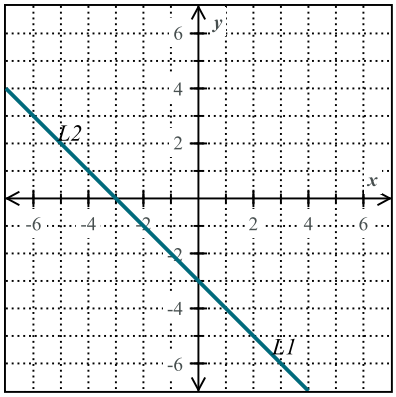
System C

L1: $y = -x - 3$

L2: $x + y = -3$



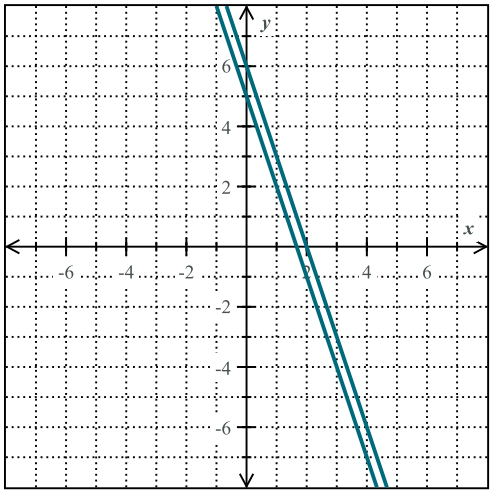
This system of equations is:

- consistent dependent

This means the system has:

- infinitely many solutions

Question 16 of 52



Solution: No solution

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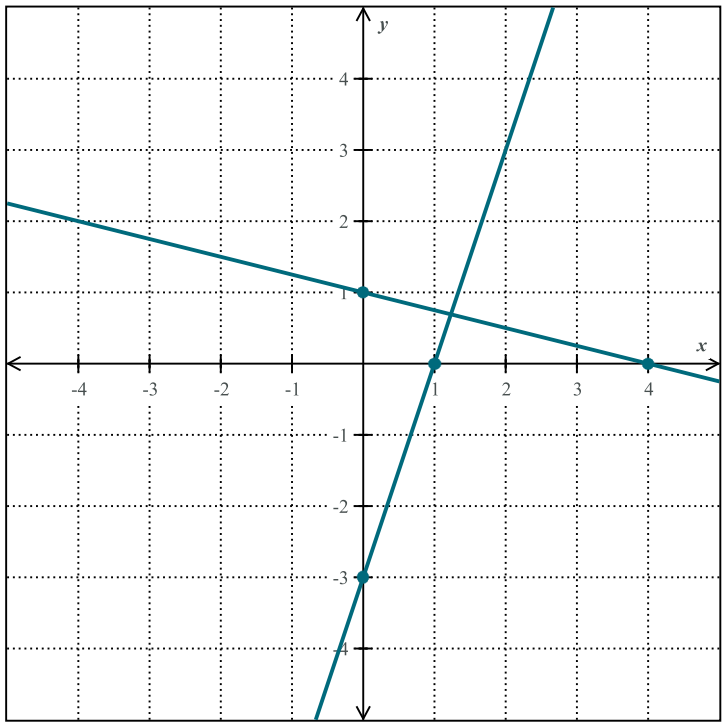
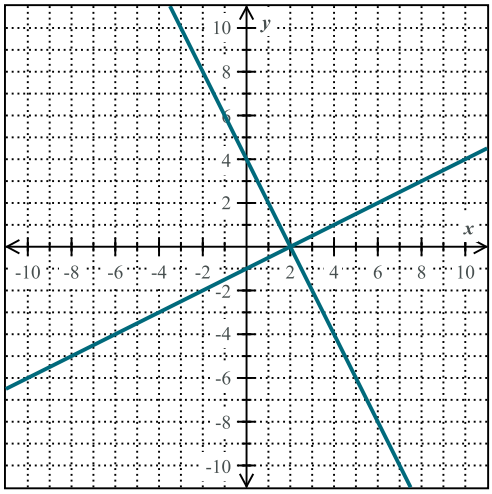
Question 17 of 52

(a)



(b) $(1.2, 0.7)$

Question 18 of 52



Solution: $(2, 0)$

Question 19 of 52

$(x, y) = (-0.44, 1.89)$

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Question 20 of 52

$(x, y) = (6.87, 0.95)$

Question 21 of 52

(a) After 250 miles driven, which car will have less gas remaining?

- ○ Tony's car
- ◉ Reuben's car

How much less gas will it have?

2 gallon(s)

(b) After how many miles will the tanks contain the same amount of gas?

150

If the number of miles driven is less than this, which car will have more gas remaining?

- ○ Tony's car
- ◉ Reuben's car

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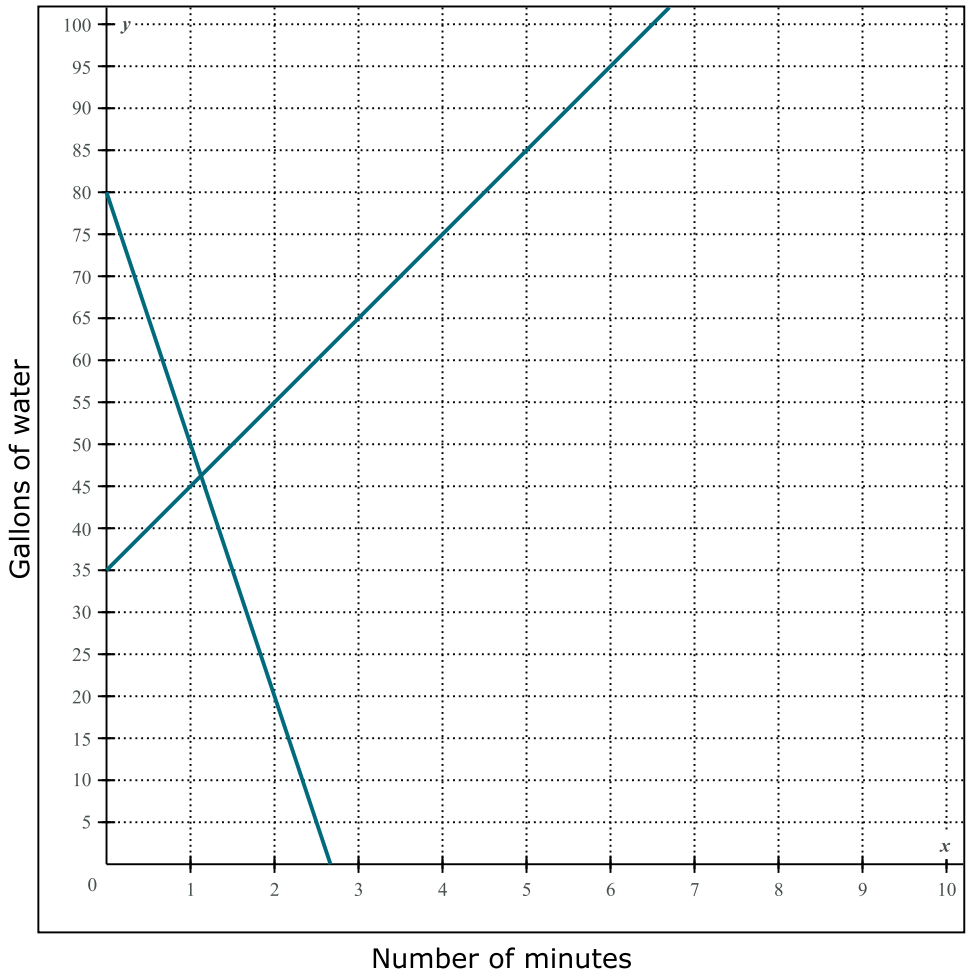
Question 22 of 52

(a)



(b) 1.1 minutes

(c) 46 gallons

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Question 23 of 52

(a) Is $(-2,\ -5)$ a solution?

- Yes
- No

(b) Is $(1,\ 2)$ a solution?

- Yes
- No

(c) Is $(-9,\ -6)$ a solution?

- Yes
- No

(d) Is $(0,\ -8)$ a solution?

- Yes
- No

Question 24 of 52

$x = 4$

$y = -1$

Question 25 of 52

$x = -10$

$y = 3$

Question 26 of 52

$x = 5$

$y = 1$

Question 27 of 52

$x = -15$

$y = -15$

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Question 28 of 52

$x = 7$
$y = 5$

Question 29 of 52

System A $y = 9x + 7$ $y = 9x + 4$	◉ The system has no solution. ○ The system has a unique solution: $(x, y) = (_____, _____)$ ○ The system has infinitely many solutions.
System B $-x - 2y = 3$ $x + 2y = -3$	○ The system has no solution. ○ The system has a unique solution: $(x, y) = (_____, _____)$ ◉ The system has infinitely many solutions.

Question 30 of 52

(a) ○ The system has no solution.

○ The system has a unique solution.

$(x, y) = (\square, \square)$

◉ The system has infinitely many solutions. They must satisfy the following equation.

$y = 3x - 6$

(b) ◉ The system has no solution.

○ The system has a unique solution.

$(x, y) = (\square, \square)$

○ The system has infinitely many solutions. They must satisfy the following equation.

$y = \square$

Question 31 of 52

Number of chemistry textbooks sold: 266

Number of biology textbooks sold: 179

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Question 32 of 52

Cost to rent each chair: $\$1.25$

Cost to rent each table: $\$8.25$

Question 33 of 52

(a) Write a system of equations that could be used to find the weight (in kilograms) of each type of box.

$$3x + 2y = 53$$

$$8x + 4y = 133$$

(b) How much does each type of box weigh (in kilograms)?

Weight of each large box: 13.5 kilograms

Weight of each small box: 6.25 kilograms

Question 34 of 52

(a)

$$5x + 3y = 32$$

$$2x + 12y = 83$$

(b) Cost for each game: $\$2.50$

Cost for each audiobook: $\$6.50$

Question 35 of 52

First car: $40\ \frac{\text{g}}{\text{km}}$

Second car: $35\ \frac{\text{g}}{\text{km}}$

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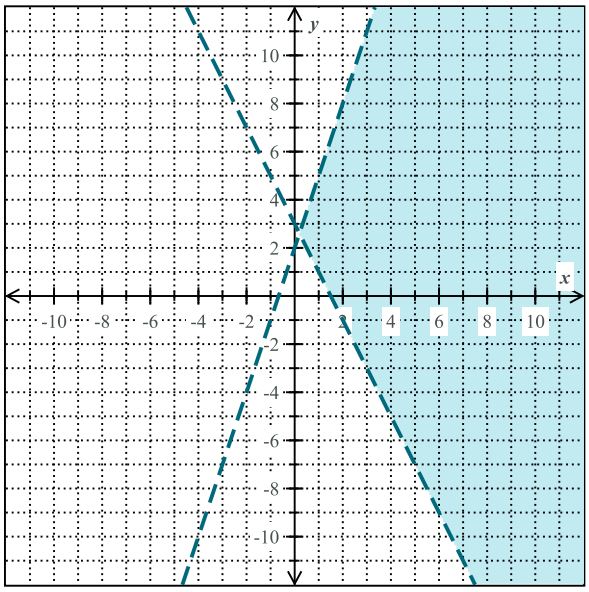
Question 36 of 52

How much of each brand should she use?

First brand: 48 milliliters

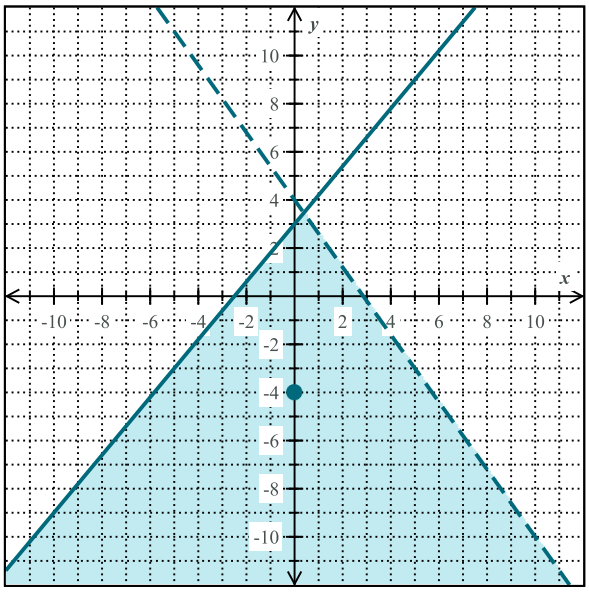
Second brand: 192 milliliters

Question 37 of 52



Question 38 of 52

(a)



(b) Point in the solution set: $(0,\ -4)$

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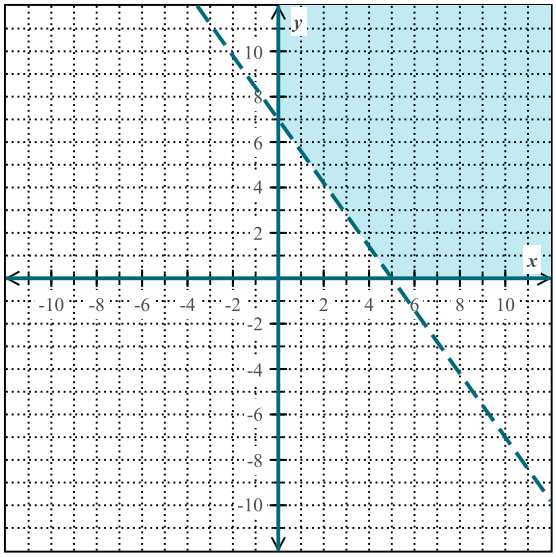
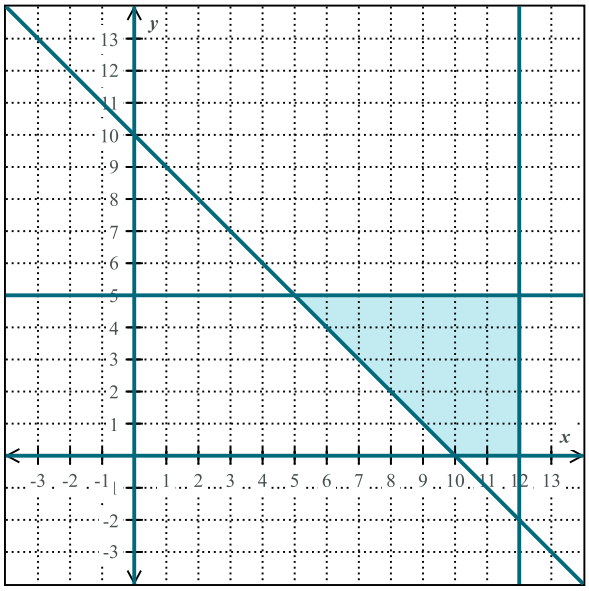
Question 39 of 52



Question 40 of 52

$3x + 4y > 93$

Question 41 of 52



Note that we could have graphed the above region without drawing the line $x = 0$.

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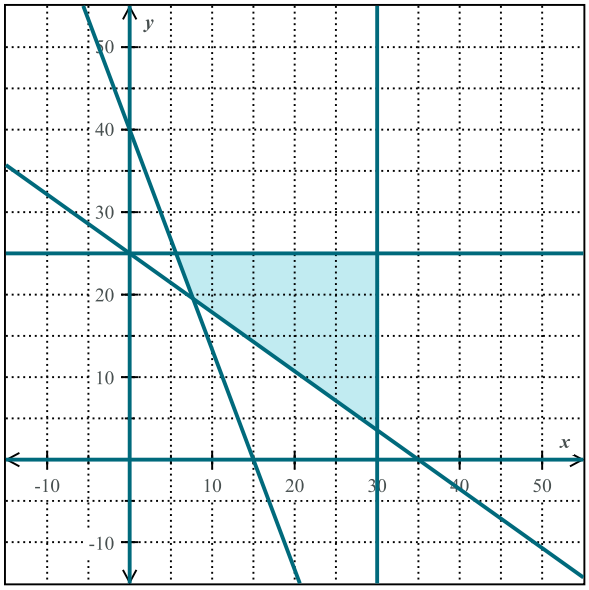
Question 42 of 52



Question 43 of 52

(a) $30x + 60y \leq 600$

(b) $x + y \geq 11$

(c)
- ○ $x > y$ because there will be more white balls than black balls in the bag.
- ◉ $x \geq 0$ and $y \geq 0$ because the number of black balls and number of white balls cannot be negative.
- ○ $x < 0$ and $y < 0$ because each ball takes up space in the bag.

(d)

Description	Possible?
8 black balls and 4 white balls	◉ Yes ○ No
6 black balls and 10 white balls	○ Yes ◉ No
12 black balls and 0 white balls	◉ Yes ○ No

Question 44 of 52

$z = 126$

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Question 45 of 52

First mine: 37 hour(s)
Second mine: 11 hour(s)

Question 46 of 52

First number:	22
Second number:	27
Third number:	88

Question 47 of 52

$x=2$

$y=-1$

$z=1$

Question 48 of 52

$x=2$
$y=5$
$z=-3$

Question 49 of 52

$x=-2$

$y=3$

$z=-2$

Question 50 of 52

How many phone calls did she receive each evening?

Number of phone calls the first evening: 19

Number of phone calls the second evening: 11

Number of phone calls the third evening: 57

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Question 51 of 52

Cost to have a coat cleaned: $\$5$

Cost to have a shirt cleaned: $\$2$

Cost to have a dress cleaned: $\$4$

Question 52 of 52

slope: -1

y-intercept: -5

equation: $y = -x - 5$

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